

OpenSSH

Objectives

After completing this lesson, you should be able to:

- Describe OpenSSH
- Describe OpenSSH configuration files
- Configure OpenSSH server and client
- Use the `ssh` command
- Use the `scp` command
- Use the `sftp` command
- Use the `ssh-keygen` command
- Describe the use of key pairs for Oracle Cloud Infrastructure instances
- Use `ssh-agent` and `ssh-add`



OpenSSH: Introduction

OpenSSH:

- Is a suite of secure network connectivity tools:
 - `ssh`: Secure shell command
 - `scp`: Secure copy command
 - `sftp`: Secure file transfer protocol (FTP) command
 - `sshd`: OpenSSH daemon
 - `ssh-keygen`: Creates ECDSA or RSA authentication keys
- Is a secure alternative to `telnet`, `rcp`, `rsh`, `rlogin`, and `ftp`
- Encrypts all communication between the client and server
- Supports the SSH2 protocol on client and server
 - SSH1 removed from server side as of Oracle Linux 7 Update 4
- Provides X11 forwarding and port forwarding



OpenSSH Configuration Files

- Global files are stored in the `/etc/ssh` directory.
- User files are stored in the `~/.ssh` directory.
- Global files include:
 - `ssh_config`: The default OpenSSH client configuration file
 - `sshd_config`: The configuration file for the `sshd` daemon
 - Various ECDSA and RSA public and private key files
 - PAM configuration file: `/etc/pam.d/sshd`
 - Configuration file for `sshd`: `/etc/sysconfig/sshd`
- User files include:
 - `config`: Overrides global `ssh_config` file
 - `known_hosts`: Contains host keys of SSH servers accessed by the user
 - Various user ECDSA and RSA public and private key files

OpenSSH Configuration

- To configure an OpenSSH server:
 - The following packages are installed by default:
 - openssh
 - openssh-server
 - Start the `sshd` daemon:

```
# systemctl start sshd
```

- Configure the service to start at boot time:

```
# systemctl enable sshd
```

- To configure an OpenSSH client:
 - The following packages are installed by default:
 - openssh
 - openssh-clients
 - There are no services to start on the client.

Using OpenSSH Utilities

- All OpenSSH utilities require a remote user account.
- The first time you connect to an OpenSSH server, the OpenSSH client prompts you to confirm that you are connected to the correct system:

```
$ ssh host03
The authenticity of host 'host03 (192.0.2.103)' can't be established.
RSA key fingerprint is ...
Are you sure you want to continue connecting (yes/no)? Yes
Warning: Permanently added 'host03,192.0,2,103' (RSA) to the list of known
hosts.
```

- The user's `~/.ssh/known_hosts` file is updated.

Using the `ssh` Command

- The `ssh` command allows you to:
 - Connect to a remote system
 - Execute a command on a remote system
- The format of the command is:

```
ssh [options] [user@]host [command]
```

- Examples:

```
$ ssh host03  
$ ssh root@host03  
$ ssh host03 ls
```

Using the `scp` Command

- Use `scp` to copy files or directories to or from a remote system.
- To copy to a remote system, the format is:

```
scp [options] local-file [user@]to-host:remote-file
```

Examples:

```
$ scp test host03:~  
$ scp test host03:~/new_test
```

- To copy from a remote system, the format is:

```
scp [options] [user@]from-host:remote-file local-file
```

Examples:

```
$ scp host03:~/new_test .  
$ scp host03:~/new_test newer_test
```


Using the `sftp` Command

- `sftp` is a secure alternative to, and is functionally the same as, `ftp`.
- The format to connect to a remote system is:

```
sftp [options] [user@]host
```

Example:

```
$ sftp host03
```

- You are presented with the `sftp>` prompt after connecting:

```
sftp>
```

- Enter `help` or `?` to display a list of `sftp` commands.
- To upload a file (copy to remote system):

```
sftp> put filename
```

- To download a file (copy from a remote system):

```
sftp> get filename
```

Using the `ssh-keygen` Command

- The `ssh-keygen` command generates authentication key pairs.
- Use the `-t` option to specify the key type. Example:

```
$ ssh-keygen -t rsa
```

- `ssh-keygen` generates two keys:
 - Private key
 - Public key
- Specify a passphrase to encrypt the private part of the key.
- To allow remote connectivity without supplying a password:
 - Copy the public key to `~/.ssh` on the remote system.
 - Do one of the following:
 - Rename the public key file `authorized_keys`.
 - Append the public key to the `authorized_keys` file on the remote system to allow connection from multiple clients.

Key Pairs for Oracle Cloud Infrastructure Instances

- Generated key pairs are required for Oracle Linux instances.
 - Authenticates remote users instead of using passwords.
- Public keys are provided when launching instances.
- Private keys are specified when connecting to instances with SSH.
- For example, to log in with a private key contained in the `id_rsa` file in the `/root/.ssh` directory:

```
# ssh -i ~/.ssh/id_rsa opc@<public-ip-address>
```

- Oracle Linux images use the default user name `opc`.
- You supply the public IP address of your instance.

Using ssh-agent

- `ssh-agent` is an authentication agent that handles passwords for SSH private keys.
 - Use `ssh-keygen` to generate authentication key pairs.
 - Provide a passphrase, for example “password”, when creating the key pairs.
 - Copy the public key to `~/.ssh/authorized_keys` on the remote system.
- To use `ssh-agent`:

```
$ exec ssh-agent $SHELL
```

- Use `ssh-add` to add the keys:

```
$ ssh-add
```

```
Enter passphrase for /home/oracle/.ssh/id_rsa: password
```

```
Identity added: /home/oracle/.ssh/id_rsa ...
```

Quiz



OpenSSH connectivity tools encrypt all network traffic, including passwords.

- a. True
- b. False



Summary

In this lesson, you should have learned how to:

- Describe OpenSSH
- Describe OpenSSH configuration files
- Configure OpenSSH server and client
- Use the `ssh` command
- Use the `scp` command
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Practices for Lesson 15: Overview

This practice covers the following topics:

- Connecting to a remote server by using `ssh`
- Configuring OpenSSH to connect without a password